Junxian Ma

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Education

Tsinghua University *2024–2028 (expected)*

Undergraduate in Computer Science and Technology | Beijing

Research Interests

My research interests include distributed LLM inference, sparse attention, GPU optimization, and algorithm–system co-design.

Projects

FusedIndexTopK *Jul–Oct 2026*

- Built a fused Indexer + Top-K operator for DeepSeek-V3.2 on H20 GPUs, avoiding writes and rereads of the dense score matrix.
- Reduced average operator latency by 3.8–9.5% relative to DeepGEMM + DeepSelect across context lengths of 16K–128K on H20 GPUs.

Tsinghua University Supercomputing Team *2026–Present*

Training Participant

MFC Performance Analysis and Optimization *Oct 2026–Present*

- Profiled the MFC multiphase flow solver on NVIDIA H100 GPUs and identified bottlenecks in WENO reconstruction and the HLLC Riemann solver.
- Evaluated device link-time optimization (LTO), analyzed stack frames, register usage, and memory access behavior, and performed targeted tuning.

OpenHPL-Ozaki *Oct 2026–Present*

- Investigating INT8 Tensor Core emulation of FP64 matrix multiplication in a single-GPU HPL project for Hopper GPUs.
- Surveying FP64 emulation methods, including Ozaki I/II and EmuGEMM, and GPU HPL implementations to identify strategies for improving HPL performance.

Internship Experience

Zhipu AI | Multimodal Algorithm Intern *Oct 2025–Feb 2026*

GLM-4.6V Series | Beijing

- Contributed to multimodal alignment and long-video understanding for GLM-4.6V, including model architecture improvements and reinforcement learning training to support those changes.
- Built large-scale multimodal data pipelines for data cleaning, filtering, and quality control.

Skills and Honors

Programming Languages: Python, C/C++, CUDA

2026 Comprehensive Excellence Scholarship, Tsinghua University
2025 Outstanding Student Cadre, Tsinghua University
2024 First Place in the Gaokao in Gansu Province
2024 First-Class Freshman Scholarship, Tsinghua University
2023 Bronze Medal, Finals of the 40th Chinese Physics Olympiad